Qihan (Blake) Wang

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Education

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
B.S. in Electrical Engineering — *Expected May 2029*

- **Relevant Coursework:** Differential Equations; Calculus III; Linear Algebra; Discrete Structures; Introduction to Electronics

Experience

CUDA-Accelerated Nuclear Pasta Research — Oct. 2026–Present
Research with Dr. Soham Pal | CUDA C++, Linux

- Implemented an FP64 CUDA kernel for all-pairs nuclear and screened Coulomb force calculations using a structure-of-arrays layout, periodic boundaries, and shifted-force cutoffs.
- Built CPU/GPU numerical validation and repeatability tests; implemented CUDA-event benchmarks with persistent device buffers and explicit host–device memory transfers.

EEG Motor-Imagery Classification and Signal Processing — 2026
Independent Research Project

- Built a Python/MNE EEG preprocessing pipeline implementing bandpass filtering, epoch segmentation, and frequency-domain analysis.
- Implemented time-, frequency-, and spatial-domain EEG features, including band power, Hjorth parameters, wavelet energy, CSP, and FBCSP; evaluated SVM, random forest, logistic regression, and shrinkage LDA classifiers.

Computer Science/Cybersecurity Research Intern — Summer 2025
University of Houston, Department of Computer Science

- Developed Python scripts to evaluate large language model reasoning accuracy and cross-domain knowledge transfer across biology, mathematics, and law.

Projects

Chronodox Strategic Simulation Platform | *React, TypeScript, Zustand, Vite, Node.js* — 2026

- Developed a browser-based multiplayer strategy simulation with a rule-based engine, procedural map generation and spatial algorithms, and an in-browser terminal interface that exposes game actions, diagnostics, and training workflows
- Implemented REST-based client–server synchronization, built a browser-based AI self-play pipelines and trained a hybrid heuristic/candidate-based gameplay agent using 200,000+ self-play samples.

Illini Solar Car | *PCB Design, KiCad, Firmware, MCUXpresso IDE* — Sept. 2026–Present

- Designed an LED/resistor PCB in KiCad and implemented potentiometer-controlled LED brightness in MCUXpresso IDE.

Activities & Leadership

CyberPatriot | *Platinum Semifinalist, 5th Place NJROTC Semifinals* — 2025

- Hardened Linux systems in virtualized environments using Bash, system administration, and cybersecurity tools.

National Program Director, Steel City Codes — 2025–2026

- Supported computer science education through curriculum development, outreach, and program coordination; created digital outreach content generating approximately **50,000 views**.

Technical Skills

Languages: C++, C, Python, Java, Bash, TypeScript, JavaScript
Platforms: CUDA, Linux, Git, virtual machines, KiCad, MCUXpresso IDE
Libraries & Frameworks: Numpy, MNE, React, Node.js, HTML, CSS